6 **(a)** From the graph, temperature at x = 17.5 cm is $\vartheta = 45^0$.

(b) **(i)** Equal since the rod is insulated.

(ii) Temperature gradient

$$= \frac{(45 - 100)(17.5)}{(-65)(17.5)} = -3.1^0\text{C cm}^{-1}$$

Note: Do not misread scales on axes. Use required precision when reading coordinate values. Try to use end points of line to obtain gradient within range.

(iii) The metal rod in Fig 6.2 is a good conductor of heat and since it is not insulated, there is heat loss through its surface. The metal rod in Fig. 6.4, on the other hand, is insulated; hence there is no heat loss through its surface. It is thus able to maintain a higher temperature at all points along its length.

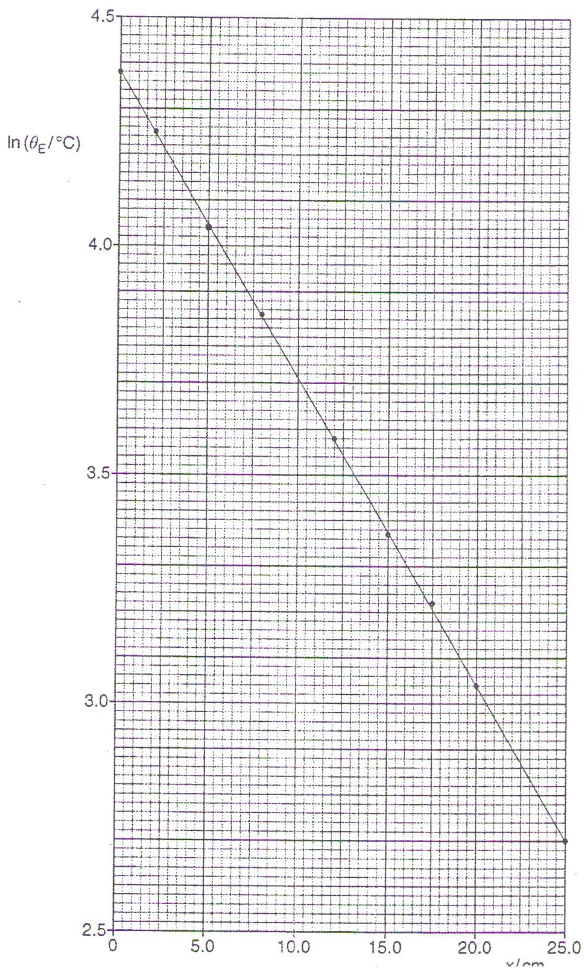
(c) When the distance is tripled, eq from 5 cm to 15 cm, the temperature did not correspondingly fall to a third from 84^0C to 28^0C but instead to 53^0C. This violates an inverse proportional mathematical relationship between temperature and distance.

(d) (i) At x=5.0 cm,

θ_E /^{0} C=77-20=57^{0}C

$$\ln\frac{\theta_E}{^0C} = \ln(57) = 4.04$$

(ii) 1.



(ii) 2. Taking log on both sides,

$$\ln\theta_E = \ln(\theta_0) - \mu x$$

which is the equation of a straight line graph of the form $y = mx + c$. Since the plotted graph is a straight line, this supports the exponential relationship between temperature and distance.

3. From the graph, the y-intercept

$\ln\theta_E = 4.38$

hence $\theta_0 = 80^0$c

Gradient $-\mu = \dfrac{(2.70-4.38)}{25.0} = -0.067\text{cm}^{-1}$

hence $\mu = -0.067\text{cm}^{-1}$.

(e) As wood is a thermal insulator, rate of heat transferred to rod will be lower hence overall lower temperature than metal rod.